

Note 6.2 Systems of Linear Differential Equations

Intro

First-Order Equations

The system

$$Y' = AY$$

where

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, Y' = \begin{pmatrix} y_1' \\ \vdots \\ y_n' \end{pmatrix}$$

is the systems of first-order linear differential equations.

And we need to solve the matrix Y .

$$Y = e^{\lambda t} \mathbf{x}$$

where λ is an eigenvalue of A is a solution, y_i is the function of t , and the $\mathbf{x}$ is the coefficient vector of $y_i(t) = x_i e^{\lambda t}$.

And any linear combination of $Y_1, Y_2, \dots, Y_n$ are all the solution of this system where $Y_1, \dots, Y_n$ with respect to $\lambda_1, \dots, \lambda_n$.

And if we know a prescribed value Y_0 when $t = 0$, the problem have a unique solution. And it is called an initial value problem.

Complex Eigenvalues

$e^{\lambda t} \mathbf{x}$ and $e^{\bar{\lambda} t} \bar{\mathbf{x}}$ are both solutions of the first-order system $Y' = AY$. And any linear combination of these solutions will be a solution. Thus we can set

$$Y_1 = \frac{1}{2}(e^{\lambda t} \mathbf{x} + e^{\bar{\lambda} t} \bar{\mathbf{x}}) = \text{Re}(e^{\lambda t} \mathbf{x})$$
$$Y_2 = \frac{1}{2}(e^{\lambda t} \mathbf{x} - e^{\bar{\lambda} t} \bar{\mathbf{x}}) = \text{Im}(e^{\lambda t} \mathbf{x})$$

and it is not necessary to find and solve $\bar{\mathbf{x}}, \bar{\lambda}$.

Higher Order Systems

If we have an m th-order system

$$Y^{(m)} = A_1 Y + A_2 Y' + \dots + A_m Y^{(m-1)}$$

where A_i is an $n \times n$ matrix. We can transform it into a first-order system by setting

$$Y_1 = Y, Y_2 = Y', \dots, Y_m = Y_{m-1}'$$

And we can use the form

$$\begin{pmatrix} Y_1' \\ Y_2' \\ \vdots \\ Y_{m-1}' \\ Y_m' \end{pmatrix} = \begin{pmatrix} O & I & O & \dots & O \\ O & O & I & \dots & O \\ \vdots & & & & \\ O & O & O & \dots & I \\ A_1 & A_2 & A_3 & \dots & A_m \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_{m-1} \\ Y_m \end{pmatrix}$$

And we require $Y, Y', \dots, Y^{(m-1)}$ take on specific values to ensure unique solution.

For the system simply in the form of $Y^{(m)} = AY$, the solution is

$$Y = e^{\delta t} \mathbf{x}$$

where δ is the m th roots of the eigenvalue of A , that is, $\delta = \sqrt[m]{\lambda}$.